MESW · SOFTWARE ENGINEERING LABORATORY

How this laboratory works this semester

First class · 2026/27

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A product. The process that produced it. People who can defend both.

This is a laboratory, not a lecture course.

Six ECTS. Thirteen weeks. A product, by a process you can show, in a team we assemble.

01

A running increment

Something works, on a tag, against the locked MVP. That is the product half.

02

A process you can open

Stories, tests, pipeline, history, retro. If we cannot open it in the tag, it did not happen.

03

Judgement you can defend

Any one of you, any review. The tool is staff. You still own the call.

Five photographs. One product.

Each review is a photograph. Early stumbles can be recovered. An empty finish cannot.

Iter 0

Open

Ready to build

Vision + locked MVP
+ a thin slice that runs

Iter 1

Move

First increment

MVP starts to move

Iter 2

Show

Product visible

Someone can use it

Iter 3

Harden

Practices show

Pipeline, tests, honesty

Iter 4

Ship

Pre-Release

Last increment +
current practice health

The calendar

2 Oct	9 Oct	16 Oct	23 Oct	30 Oct	6 Nov	13 Nov	20 Nov	27 Nov	4 Dec	11 Dec	18 Dec
Kick-off	**Project Assignment**	**Product Management**	**ITERATION #1**		**ITERATION #2**		**ITERATION #3**		**ITERATION #4**		**Wrap-Up**
• Getting acquainted in the Lab. • Group Formation (roles and members) • Overview of full semester work plan.	• Review and discussion of assignments. • Projects assignment to teams. • Setting up of project/process support infrastructure. .	• Vision • MVP • Vertical Prototype	Iteration 1 : Start	FEUP Academic Break Iteration 1 : ongoing	• Iteration 1: End (**review**, retrospective) • Iteration 2 : Start	Iteration 2 : ongoing	• Iteration 2: End (**review**, retrospective) • Iteration 3 : Start	Iteration 3 : ongoing	• Iteration 3: End (**review** and retrospective) • Iteration 4: Start	Iteration 4: ongoing	• Iteration 4: End (review and retrospective) • Project Wrap-Up • Final Release >>>
			SPRINT REVIEW		**SPRINT REVIEW**		**SPRINT REVIEW**		**SPRINT REVIEW**		**SPRINT REVIEW**

You do not pick your team.

Four or five people. Mixed backgrounds. That is the professional lesson.

We assemble the teams

Heterogeneous on purpose. You land on a predefined group, as you will at work.

Membership will drift

People drop, arrive late, miss weeks. Scope is cut in public. Marks attach to people.

Your mark is not only the team’s

Team review, then plus or minus at most two values from how you defend the increment.

One room. You arrange it.

Every laboratory you may rebuild the furniture into a cluster for your team. Put it back when you leave.

Clusters, not rows

Tables and chairs are yours to configure. Face each other. A lecture layout is a bad place to build a product.

Lift. Do not drag.

Pick tables and chairs up. Do not pull them across the floor. Adjacent rooms have classes in session.

Quiet is part of the lab

Talk to your team. Do not talk to the corridor. If the room is loud, the next class hears it.

Dream on the Vision. Ship the MVP.

Locked at the end of Iteration 0. After that, the Vision is a horizon.

VISION

The ultimate product.

- Go wide.
- Stay ambitious. This is permission to think.
- It may grow.
- It is not what we mark after week 4.

MVP · THE CONTRACT

The product we review.

- Who it is for.
- A handful of MVP outcomes (user-visible things the product must do) — typically three to five; more than six is the Vision leaking.
- What is out.
- A definition of done.
- A thin vertical slice that runs — UI to data. A mock-up is not a slice.

You may cut it in public or expand from the Vision once it is accepted. You may not quietly replace it.

An outcome is a thing the user can do.

Not a screen. Not a user story. Typically three to five; more than six is the Vision leaking.

CAMPUS DATES

- See my evaluation dates this semester.
- Record that I cannot attend one.
- Get one reminder before a review.

STOCK ROOM

- Add an item to stock.
- Record a stock-out.
- See what is below a threshold.

PICK-UP COUNTER

- Place an order for pick-up.
- See whether it is ready.
- Mark it collected.

A story is done when this is true.

House minimum. Teams may add lines that can fail. They may not delete these five.

1	It runs on the tag — not only on one laptop.
2	It matches an MVP outcome (a user-visible thing the product must do), or an explicit public cut.
3	A check exists that would fail if the change broke. Iter 0: a walkthrough is enough. From Iter 1, automation must appear and grow.
4	The change is in history on the tag — not a Friday dump.
5	Any of you could open it. If only the author can, it is not team-done.

A row of tens does not produce 18+.

Weather on the repository. The mark in the room.

DASHBOARD

How you are faring

- After each review you get 0–10 notes on practices: vision, stories, tests, retro, increment, servers, CI, history, AI log.
- They tell you where to attend. 8–10 means used and defendable — not rich, not numerous.

SPRINT REVIEW

The official mark

- Twelve to fifteen minutes, standing at the machine. Increment versus locked MVP. History. We pick students. One place a tool was wrong.
- Marks of 18+ are given here. Not on the wiki.

Ten lines. Each iteration is a photograph.

Weather, not the mark. You choose what to invest when.

LIVING

May look better in the next photograph

- Vision · Stories · Acceptance tests · Unit tests · Staging / production · CI/CD · Repository health
- An early 4 is a signal, not a sentence. Repair shows later.

BOUND

Finished when the review ends

- Sprint retrospective · Product increment · AI critique
- Cannot be rewritten in December. No official process calendar — later photographs show the cost. Tens still do not produce 18+.

Prepare as if it will be you.

Product and process, every time. It might be you again.

Any one of you may have to speak for this increment — a story, a test, a pipeline, a page, a decision, or a commit.

02 min

It runs

Point at the MVP. No slide deck as the object.

06 min

We probe

One product probe and one process probe: history, pipeline, or a retro action in the compare.

04 min

You defend

We pick a student(s). A team artefact. Where the tool failed.

03 min

We close

Accept, accept with debt, or a different product.

If it is not linked, it does not exist.

Tag iter-N and CLAIM.md on that tag. We will not hunt the repository.

`docs/iterations/N/CLAIM.md`

Permalinks we follow: this tag · compare with the last tag · production · staging · pipeline · MVP · Vision · stories in this increment · how to run the tests · retrospective · AI log · team artefacts someone will be asked to open.

- A dead link scores that line at most 4.
- Missing tag or claim: you may still demo; the team mark for that iteration is capped at 10.
- Iteration 0 is a warning shot if the slice runs.

DO NOT

- Write a score on the claim.
- Ask us to browse.
- Point at a wiki page that is not on the tag.
- Treat a Friday dump as history.

How a number is born

The room wins. A handsome wiki cannot launder a dead demo.

< 10	Nothing real to run, or the tag contradicts the demo.
10–13	Something runs. MVP still foggy. Only one person can speak.
14–16	Increment matches the locked MVP. Debt is named. Two people can answer.
17–18	Operable. A break is caught. Hot-seat is real. AI miss is specific.
19–20	The above, plus a rejected design, and production without a script.

Use it as staff. Do not send it to the review.

Approved tools only. No sensitive data in unapproved systems.

GREEN

Build

Code, tests, CI, boilerplate, docs. Later expected. You must be able to defend the result.

YELLOW

Plan

Stories and design drafts. AI as staff engineer. An unreviewed generated plan is not the team plan.

RED

The room

Sprint review, hot-seat, live demo. No copilots. No generated answers read off a laptop.

No commits is not “did not work.”

“I coordinated,” with nothing we can open in this tag, is not work we can mark above 4.

THE AI LOG

- One group disclosure per iteration.
- Name the tools.
- Show one place the model was wrong, and what you changed.
- Theatrical “we used ChatGPT” is 1–4.

WHAT YOU CAN DEFEND

A commit. A test. A story. A pipeline. A page. A retro action visible in the compare. An AI miss plus the edit.

Ask yourself before the review:

- if they call my name, what do I open?
- Is there something I can open, that I can’t defend?

Ordinary labs are not compulsory for TE.

The five reviews are. They are evaluation events. Dates on the calender

The room is the default

A TE who cannot be in the building that afternoon may join on video for that review only. Request it before the review. Not a semester mode.

Same hour, same draw

Camera on for the whole team slot. You will not be told if it is you. The review does not move.

The building is the easier place

The product is on the machine in the room. Defending it from a kitchen is harder. Video is the exception, not the advantage.

No private hour

A request to be remote from now on, or to sit at another time, is a request to miss the examination.

You do not need assigned roles.

You do need visible work, a shared clock, and a way to disagree.

Make this week’s work visible

Who is on which task until the next review — not a permanent title. Artefacts are the team’s. If it is not in the tag, it is not done.

One conversation, not five chats

Decisions live in the repository. Private messages do not survive a dropout. English is the working language of the team.

Publish when you can actually work

Jobs, time zones, other courses. A silent calendar is how resentment starts. Cut scope in public when capacity drops.

Difference is the design, not a defect.

You were assembled that way on purpose — nationality, experience, first degree.

DISAGREEMENT IS WORK

Argue the product and the process early, and small. Silence until the review is not harmony. It is a surprise we will mark.

NO DEFAULT EXPERT

Passport, bachelor, or years of code do not appoint the architect. The person who can open the artefact owns the explanation.

NOTIFY US EARLY

If someone vanishes or the team is stuck, tell us while there is still a sprint. Covering for a ghost builds a weak seat, not loyalty.

Start Iteration 0.

Theme. Vision. MVP statement. A slice that runs. Claim sheet on the first tag.

We will not

— Run a detector as proof.

— Give a parallel project or a written substitute for a review.

— Reopen a closed increment because the December wiki is handsome.

— Treat a row of dashboard 10s as an 18.

Review dates · [23 Oct 6 Nov 20 Nov 4 Dec] Recovery slot · [TBD]

The review examines the team. Any one of you may have to speak for the increment.
Prepare as if it will be you. It might be you again next time.

One week. Three ideas. One yes.

The Vision may still grow. The MVP is not locked until Iteration 0 ends.

Bring three viable ideas

A user you can play. A slice that can run in this lab. No secret employer data, no hardware as the core, no platform. One page in your head is enough. No slide deck.

Pitch them in week 2

Five minutes. We listen for the team as much as for the idea. Yes, yes-with-warning, or not this one.

Ten teams, ten themes

Everyone is in this room. No two teams may share a product. Collisions we settle here. The other turma is not a separate catalogue.

A warning now; a veto if it cannot live here

If all three fail, one replacement by the next lab — not a semester of shopping. After the yes, build toward the lock.

Questions.

Ask now. The rules do not get clearer in week 4.

You may already be wondering

— What if all three ideas are vetoed?

— What if I cannot be in the room for a review?

— What do we actually bring next week?

— Anything else. The hot-seat is not this afternoon.

Nuno Flores · Jorge Melegati

One laboratory. Ten teams. Prepare as if it will be you.